ECON 1020: Principles of Microeconomics

Homework 6: Chapters 10–11

Externalities; Public Goods and Common Resources

Due on Monday, November 9 at 12:20 PM

Submissions are accepted either online through NTU COOL or as a hard copy at your recitation session.

Part A: Externalities

1. In a hypothetical competitive market for paper, marginal benefit is $MB = 100 - Q$ and marginal private cost is $MPC = 20 + Q$. Quantity Q is measured in tons per day; prices, benefits, and costs per ton are measured in NT$1,000. Each ton creates external damage of 20 in the same units. There are no external benefits.

   (a) Calculate the unregulated equilibrium quantity and price. Write the equation for marginal social cost and calculate the efficient quantity.

   (b) Draw and label MB, MPC, and MSC. Mark both quantities and shade the deadweight loss from unregulated production. Calculate that loss in NT$ per day.

   (c) Find the corrective tax per ton. Calculate the price paid by buyers, the price kept by sellers, and tax revenue in NT$ per day.

   (d) Use the deadweight loss in part (b) to state how much the corrective tax raises total surplus. Explain why surplus rises even though trade falls.

2. A hypothetical market supplies fire extinguishers. Each extinguisher benefits its owner and can also prevent a fire from spreading to neighbors. Owners consider only their own benefit. There are no external production costs, demand slopes downward, and supply slopes upward.

   (a) Explain why there is a positive externality. Distinguish the owner's private benefit from the external benefit.

   (b) Draw the curves for marginal private benefit, marginal social benefit, and marginal cost. Mark the market and efficient quantities, and explain why they differ.

   (c) Suppose the marginal external benefit is a constant NT$600 per extinguisher. State a corrective policy and explain how it changes incentives. Does the external benefit justify an unlimited subsidy?

3. A musician values an evening practice session at NT$1,200. The noise causes NT$2,000 of harm to one neighbor. There are no other effects.

   (a) If the musician has the right to practice and bargaining is costless, identify the efficient outcome and the range of payments that would make both parties strictly better off than practicing.

   (b) Would the efficient outcome change if the neighbor had the right to quiet? If the musician has the right to practice, explain how a negotiation cost of NT$1,000 could prevent an agreement. How might a social norm against evening noise reduce the externality without a payment?

4. Two factories each initially emit 80 tons of the same pollutant per month. Each ton causes the same harm regardless of its source. Factory A can reduce emissions at a constant cost of NT$2,000 per ton; Factory B's cost is NT$5,000 per ton. These costs apply to all reductions up to 80 tons per firm. The target is total emissions of 100 tons per month.

   (a) If each factory must reduce emissions by 30 tons, calculate each factory's cost and the total cost.

   (b) Instead, each factory receives 50 tradable permits. Each permit allows one ton of emissions. Find the least-cost allocation of reductions, final emissions, and permit trades needed to reach that allocation.

   (c) Suppose the factories negotiate the permit trade in part (b) at NT$3,000 per permit. Calculate each factory's net cost after trading, including permit payments or receipts. Calculate the combined cost and the savings relative to part (a). Explain why permit payments are not an additional social cost.

   (d) A student says, "A firm given free permits has no reason to reduce emissions." Explain the error. Also explain which variable a fixed emissions tax controls directly and which variable a binding permit cap controls directly. If factories can adopt cleaner technology without changing output, why does a tax on measured emissions reward this change while an output tax does not?

Part B: Public Goods and Common Resources

5. Classify each good using excludability and rivalry. Explain the access and use conditions behind each answer.

   (a) Classify a sandwich sold by a shop and a subscription video stream with ample capacity.

   (b) An untolled road is initially uncongested but becomes crowded at rush hour. Explain how its classification changes. Then explain how an enforceable toll changes excludability in each case.

   (c) A government distributes meals and funds openly available basic research. Are both outputs public goods? Explain. How can a patent change the excludability of an invention?

6. A neighborhood has 100 residents. Each values a proposed warning system at NT$2,000, and all can hear it without reducing others' benefit. Nonpayers cannot be excluded. The full cost is NT$120,000. These hypothetical benefits and costs refer to the same period, and there are no other effects or collection costs.

   (a) Calculate total benefit and net social benefit. Should the system be provided on efficiency grounds?

   (b) Voluntary contributions total only NT$50,000. Explain how this can happen even if every resident values the project at NT$2,000. Calculate an equal compulsory payment that would cover the cost and each resident's net benefit under that payment.

   (c) The 100 residents are divided into three nonoverlapping groups, A, B, and C. The system can be upgraded. The table gives each group's total willingness to pay for each additional upgrade, in NT$. All groups enjoy each upgrade together. Upgrades must be installed in order and each costs NT$30,000. Calculate the efficient number of upgrades and their combined net benefit.

Additional upgrade	Group A	Group B	Group C
First	20,000	15,000	10,000
Second	15,000	10,000	8,000
Third	8,000	6,000	4,000

   (d) Explain why surveying willingness to pay may be difficult. If benefits differ across residents, does positive total net benefit guarantee that every resident gains under an equal payment? Explain.

7. Consider a hypothetical lake with unrestricted fishing. One additional fishing trip brings NT$6,000 of expected revenue and costs NT$4,000 in fuel, equipment use, and the fisher's time. It also reduces other fishers' current and future net returns by NT$3,000. All effects are included in these figures.

   (a) Calculate the fisher's private net benefit and the trip's net benefit to society. Explain why unrestricted access can lead to excessive fishing.

   (b) Find a fee equal to the marginal external cost. Would this fisher still make the trip? Explain using the private net benefit after the fee.

   (c) Propose a second way to address overfishing. Explain how it changes incentives or constrains use, and identify one monitoring or enforcement problem.

   (d) In a short paragraph, compare the warning system in Question 6 with the fish stock here. Classify each using excludability and rivalry, and explain why one problem concerns financing provision and the other concerns limiting use.

8. A hypothetical road-safety project costs NT$600,000. It reduces each of 2,000 residents' lifetime fatality risk by 1/20,000. Each resident is willing to pay NT$400 for this reduction in their own risk. All benefits and costs refer to the same period, and there are no other effects. Use the Chapter 11 example "Valuing a Safer Intersection" as a guide.

   (a) Calculate total willingness to pay, net social benefit, and the expected number of deaths prevented. Should the project be undertaken on efficiency grounds?

   (b) Calculate the implied value of a statistical life. Explain why this measure is not the payment a particular person would accept in exchange for certain death.